



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year 2024– 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3491 & Artificial Intelligence & Machine Learning

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Uninformed Search Algorithms
- **Course Outcome:** CO1
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

Uninformed search strategies are fundamental in Artificial Intelligence, as they provide a systematic way to explore problem spaces without prior domain knowledge. A crossword puzzle activity is an effective tool to assess students' understanding of uninformed search algorithms and their applications. It encourages students to recall key concepts such as search space, goal test, path cost, and exploration strategies in an engaging and interactive manner. By solving crossword clues related to different search algorithms, their characteristics, advantages, and limitations, students can reinforce their conceptual knowledge and terminology.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- A **crossword puzzle** is an engaging and interactive word game that motivates students to recall concepts and derive answers.
- Unlike traditional multiple-choice tests, it stimulates curiosity and enhances learning through problem-solving.
- The activity was conducted in a classroom for a duration of 10 minutes. The students were given 10 questions as a crossword puzzle in the topic of uninformed search algorithms. They have to think and write the answers in the given puzzle.
- The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in Figure 1 & Figure 2. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	3	2	1	1	2	2

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	1	2	2
Justification for correlation	The students apply the basic engineering knowledge to understand the concept of search algorithms	The students analyze the given questions	The students find the solutions for the given question.	The students work in a team to find solutions for the given question.	The students communicate with their team members and explain the reason for providing particular solution for the question.	The students apply the knowledge of search algorithms to solve the question.

- Images / Screenshot of the practice:

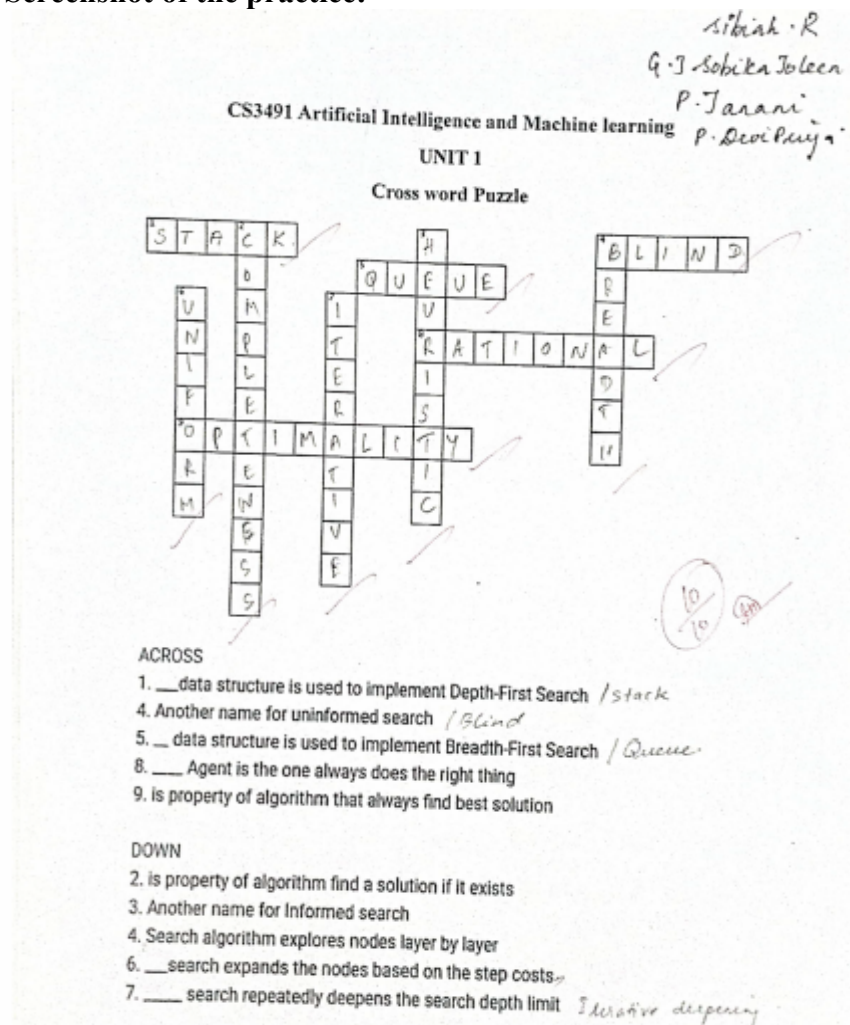


Figure 1: Sample Answers sheet shared by Sibiah. R, Sobika Joleen G. J. S, Janani. P and Devipriya.P

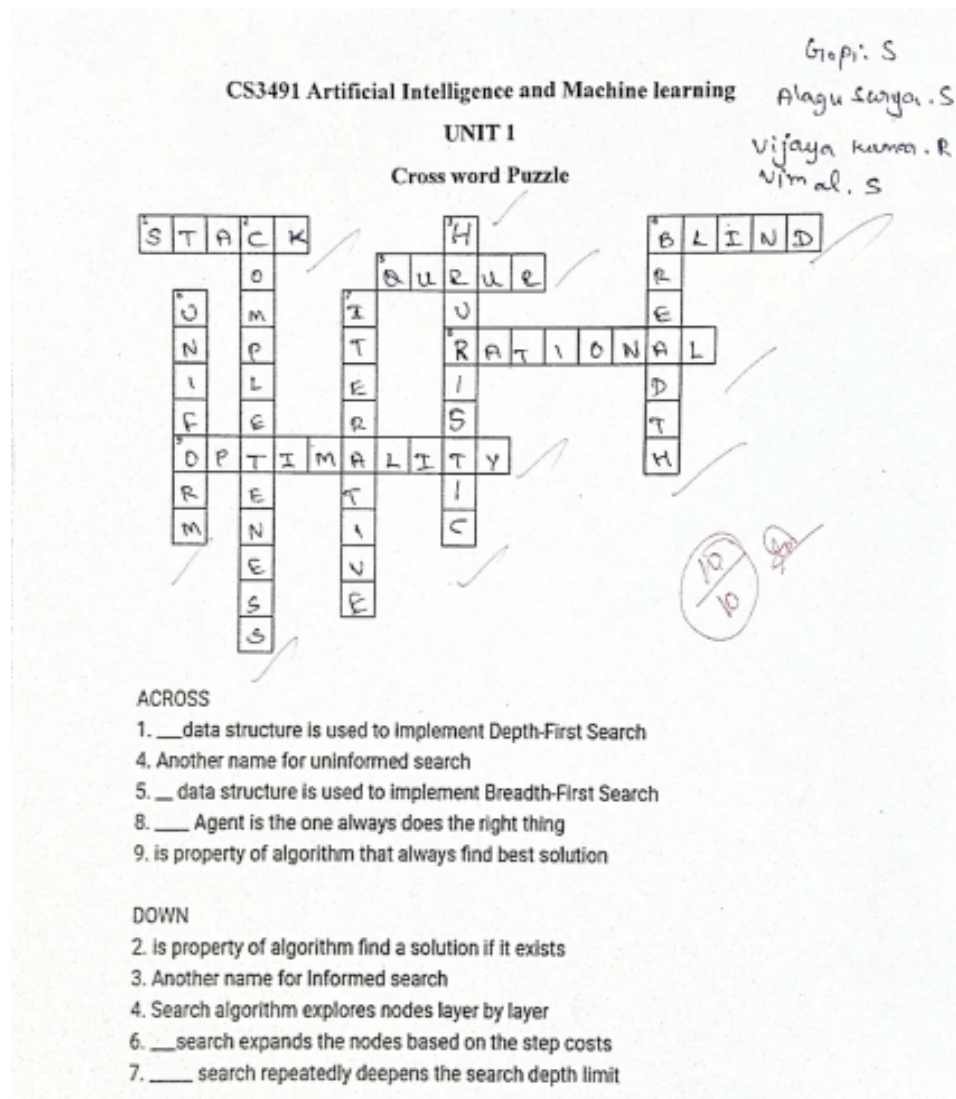


Figure 2: Sample Answers sheet shared by Gopi. S, Alagu Suriya. S, Vijaya Kumar. S, and Vimal. S

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Certain students find the puzzle straightforward and can solve it without much difficulty.
 - Few students need extra support and explanations to grasp the searching techniques represented in an uninformed search.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to enhance their ability to evaluate uninformed search strategies. The process of searching for answers and clues mirrors the steps involved in uninformed search algorithms, improving their understanding of search methods and their applications. This activity also reinforces the concept of exploring all possible options to find a solution, boosting the students' overall comprehension of the topic and strengthening their analytical skills.

❖ ***Challenges faced in implementation:***

- Some students struggle to connect the clues with the appropriate answers so they were unable to complete it within the allotted time.
- Few of them face difficulties when collaborating with team members, some get confused with their own answers and their team member answers.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ <https://crosswordlabs.com/>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2_GM_CS3353_Pointers_CWP.pdf

Signature of Faculty Member

HOD